Week 1

Team project update/progress

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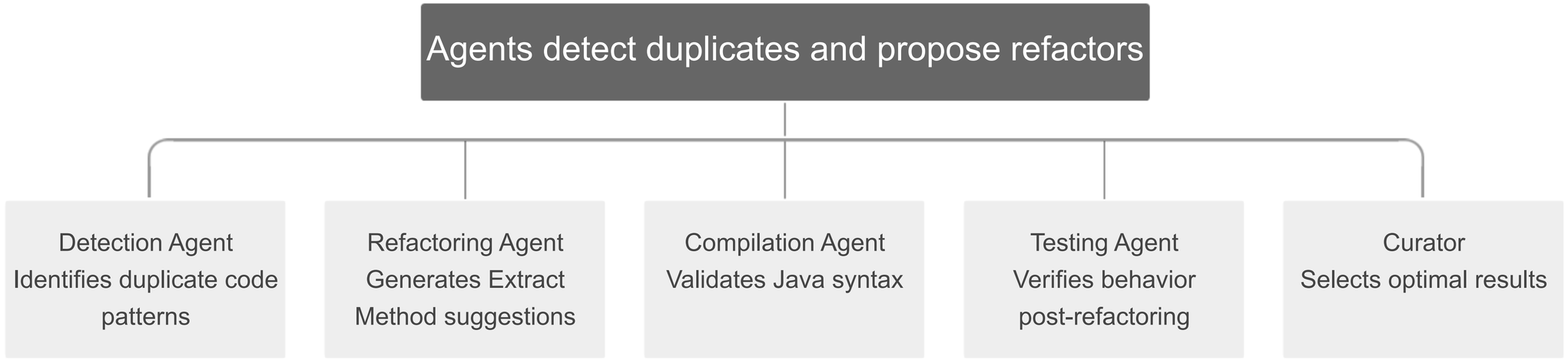
Weekly Progress Summary

GitHub	Team familiarized with basic concepts
Documentation	Reviewed and analyzed Code/PDF documentation
Architecture	Studied agent roles and system architecture
Onboarding	Team discussed about the UI design scope and objectives
UI Design	Explored design options and created prototypes
Selection	Compared prototypes and selected final design

GitHub Familiarization

Repository	Centralized file and code management
Version Control	Change tracking and version history
Task Division	Issue-based work assignment and tracking
Peer Review	Pull request workflow for code review
Git Commands	clone, pull, add, commit, push, branch, PR

Multi-Agent System Architecture



Detection Agent Workflow

1. On Paste: Checks for clones

Detection agent (3 panelists) analyzes the rest of the code in comparison to what was pasted and try to find clone ranges. If no clones are identified then no further action is taken.

2. Curator

Merges each of the panelists' outputs and returns status, identified clone ranges, and refactoring type.

Panelists

Each one is an independent LLM functioning as a code reviewer. 3 panelists total.

LLMs Used

GPT 5.2, Claude Opus 4.6, Gemini 3 Pro, DeepSeek V3.2; RAG used to augment their performance

IF LLMs fail, PSI*-fallback used

Refactoring Agent Workflow

1. Extract-Method Refactoring

Agent utilizes clone ranges provided by the detection agent as its 3 panelists each propose a refactored piece of code to replace the clone.

2. Curation

The curator analyzes each of the panelists' proposed code and picks the strongest output.

3. Test

Each one is an independent LLM functioning as a code reviewer. 3 panelists total.

LLMs Used

GPT 5.2, Claude Opus 4.6, Gemini 3 Pro, DeepSeek V3.2; RAG used to augment their performance

Usefulness Agent Workflow

1. Checks for Changes

The agent will make a check that changes have been made to the code in some capacity

2. Checks for refactoring

The agent will ensure that refactoring has occurred

3. Function Check

The agent will check that the duplicate code has been properly refactored into a function

4. Implementation Check

Finally, the agent will ensure that the refactoring has been properly implemented and that duplicate code does not remain

Compilation and Testing Agents

Compilation Agent

Simply ensures that the newly refactored code still compiles without errors

Testing Agent

Runs multiple test cases to make sure that the refactoring had no effects on the code’s functionality

Target Audience & Requirements

Target Users

Java developers and students seeking code quality improvement

Tool Characteristics

Clear, fast, and trustworthy with minimal workflow disruption

Detection Display

Show detected code with file location and line numbers

Change Visualization

Display proposed refactoring changes with compilation and testing validation status

User Actions

Apply refactoring, Edit suggestions, View Diff comparison, or Reject changes

The tool empowers developers to improve code quality efficiently while maintaining full control

Prototype Evaluation and Selection

Prototype Creation

Team created multiple design prototypes for side-by-side comparison

Clarity Assessment

Evaluated usability and visual clarity across all design options

Detail Communication

Assessed how well each design communicates and displays information

Final Selection

Chose design that best displays clone type, location, diff preview, and validation

Prototype: IntelliJ Inline Refactoring Assistant

- Follows the professor's structure: header, explanation, diff view, metadata, action bar
- Designed to feel familiar inside IntelliJ
- Adds AI Trust/Explainability to increase developer confidence
- Keeps the developer in control before applying the refactoring

Roadmap and Next Steps

Design Refinement

Polish and finalize the selected prototype design

GitHub Organization

Start utilizing it and working on branches

UI Implementation

Advance front-end development of the tool interface

Progress Tracking

Establish a website to monitor and communicate project progress